

Jaloliddin Ismailov

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EDUCATION

Eötvös Loránd University

BSc in Computer Science (GPA: 4.6/5.0; Member of Central Asian Society)

Budapest, HU

Expected: Jun 2028

EXPERIENCE

BME Solar Boat Team

Software Engineer

Budapest, HU

Feb 2026 – Present

- Developing safety-critical firmware using C and FreeRTOS on ESP32 microcontrollers to guarantee reliable throttle operation and real-time system stability during competitive races.
- Implemented bit-level error detection and scaled ADC sensor telemetry using Git to accurately monitor real-time performance metrics and prevent critical field failures.

CERN

Research Fellow at LHCb

Remote / Geneva, CH

Nov 2025 – Jan 2026

- Developed muon track reconstruction algorithms using object-oriented C++ (CUDA) on Linux to deliver highly accurate data models for experimental particle physics research.
- Automated software dependency management and CI/CD pipelines to optimize the backend development workflow and enable scalable data analysis workflows for physicists.

PROJECTS

Assistive Smart Glasses

Hardware & Firmware Developer

BME Robotics Hackathon

Jun 4 – Jun 6, 2026

- Engineered firmware and hardware architecture with C++ and PlatformIO on an ESP32-CAM microcontroller to transmit real-time telemetry data for active navigation assistance.
- [\[Video Demo\]](#)

Flood-Risk Monitoring System

Environmental Engineer

CASSINI Hackathon Budapest

Apr 24 – Apr 26, 2026

- Built a geographic mapping system using Python, GeoPandas, and REST APIs within an Agile team of three to provide clear data visualizations that help non-technical stakeholders coordinate disaster responses.
- [\[Video Demo\]](#)

EXTRACURRICULARS

European Young Engineers

Coordinator

Remote

Mar 2026 – Present

- Analysing the intersection of AI and regional climate policies across Hungary, Austria, and Slovakia to facilitate external government relations and regional strategic planning.

FLEX Program (U.S. Department of State)

Exchange Student at Prior Lake High School

Minnesota, USA

Sep 2023 – Mar 2024

- Completed over 50 hours of structured community service and collaborated alongside the International Student's Club to foster cross-cultural communication and relationship-building frameworks.